Chapter 6. Inference about Distributions

Means, paired and two-sample comparisons, variances

Contents

6.1 Inference about a population mean **2**

6.2 Comparing means by paired comparison **4**

6.3 Comparing means with two independent samples **6**

6.3.1 Large samples 6

6.3.2 Normal populations: Welch's approximation 7

6.3.3 Normal populations with a common variance 8

6.4 Inference about population variances **10**

6.4.1 One variance 10

6.4.2 Ratio of two variances 11

6.1 Inference about a population mean

Motivation. With σ unknown, replacing it by S leads to t intervals and t tests.

Replacing σ by S in $\frac{\bar{X}-\mu}{\sigma/\sqrt{n}}$ is **studentizing**. For a random sample from $N(\mu,\sigma^2)$, Theorem 4-10 gives

$$\frac{\bar{X}-\mu}{S/\sqrt{n}} \sim t(n-1).$$

Hence

$$P\Big(-t_{\alpha/2}(n-1) \le \frac{\bar{X}-\mu}{S/\sqrt{n}} \le t_{\alpha/2}(n-1)\Big) = 1-\alpha, \quad \text{i.e.}$$
$$P\Big(\bar{X} - t_{\alpha/2}(n-1)\frac{S}{\sqrt{n}} \le \mu \le \bar{X} + t_{\alpha/2}(n-1)\frac{S}{\sqrt{n}}\Big) = 1-\alpha.$$

Definition (Confidence interval for μ (normal population, σ unknown))

A $100(1-\alpha)\%$ confidence interval for μ is

$$\Big(\bar{x} - t_{\alpha/2}(n-1)\frac{s}{\sqrt{n}},\ \bar{x} + t_{\alpha/2}(n-1)\frac{s}{\sqrt{n}}\Big).$$

For large n the normality assumption can be relaxed: by the CLT and since $t(n-1)$ approaches $N(0,1)$, the quantile $t_{\alpha/2}(n-1)$ may be replaced by $z_{\alpha/2}$.

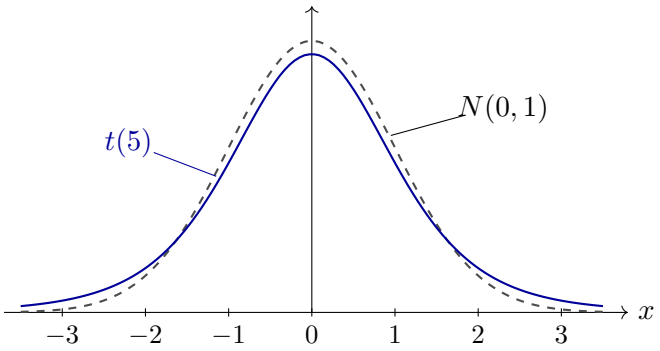


Figure 1: The $t(5)$ density (solid) has heavier tails than $N(0,1)$ (dashed).

Example 6-1 (Trees per 100 m^2)

A forest manager counted trees in 50 randomly chosen 100 m^2 plots of a 9 km^2 forest:

43	28	23	18	46	39	40	43	36	25
26	28	20	22	17	26	17	12	16	35
37	12	32	47	20	25	13	20	46	23
37	29	23	37	34	27	33	29	24	19
37	23	37	28	27	35	30	26	33	35

Find a 99% confidence interval for the mean number μ of trees per 100 m^2.
Solution. With $n = 50$,

$$\bar{x} = \frac{43+28+\cdots+35}{50} = 28.76, \qquad s = \sqrt{\frac{(43-28.76)^2+\cdots+(35-28.76)^2}{49}} = 9.146.$$

The normal quantile plot is close to a line, so a normal population is reasonable. The table lacks 49 degrees of freedom; using the nearest value $t_{0.005}(50) = 2.678$,

$$\bar{x} \pm t_{0.005}(49)\frac{s}{\sqrt{n}} \approx 28.76 \pm 2.678 \times \frac{9.146}{\sqrt{50}} = 28.76 \pm 3.464,$$

so the 99% confidence interval is (25.296, 32.223).

Testing with the studentized mean is the t **test**.

Definition (t test for μ (normal population, σ unknown))

For $H_0 : \mu = \mu_0$ use $T = \dfrac{\bar{X} - \mu_0}{S/\sqrt{n}}$, which is $t(n-1)$ under H_0. With observed value t:

Alternative	P-value	Rejection region at level α
(a) $H_1 : \mu > \mu_0$	$P(T \geq t)$	$t \geq t_\alpha(n-1)$
(b) $H_1 : \mu < \mu_0$	$P(T \leq t)$	$t \leq -t_\alpha(n-1)$
(c) $H_1 : \mu \neq \mu_0$	$P(\lvert T\rvert \geq \lvert t\rvert)$	$\lvert t\rvert \geq t_{\alpha/2}(n-1)$

(The rejection regions have Type I error exactly α because $T \sim t(n-1)$ when $\mu = \mu_0$; e.g. in (c), $P(|T| \geq t_{\alpha/2}(n-1)) = 2 \cdot \frac{\alpha}{2}$ by symmetry.)

Example 6-2 (Light bulb lifetimes)

Current bulbs last 1950 hours on average. Nine prototypes of a new bulb lasted (hours)

2000	1975	1900	2000	1950	1850	1950	2100	1975

(a) State suitable hypotheses.
(b) Assuming normal lifetimes, test them at level 5%.
(c) Find the P-value of (b).
Solution.
(a) $H_0 : \mu = 1950$, $H_1 : \mu > 1950$.
(b) $\bar{x} = 1966.7$, $s = 69.6$, so $t = \dfrac{1966.7 - 1950}{69.6/\sqrt{9}} = 0.718$. The rejection region is $t \geq t_{0.05}(8) = 1.860$; since $0.718 < 1.860$, H_0 is not rejected at 5%.
(c) $P = P(T \geq 0.718)$, $T \sim t(8)$: between the upper 25% (0.706) and 20% (0.889) points, so $0.20 < P < 0.25$ (software: 0.247). The evidence is weak.

The Q–Q plot looks roughly linear, but with so few observations the check is weak; subject knowledge or Chapter 10's nonparametric methods may help.

Example 6-3 (Cholesterol of immigrant women)

The national mean cholesterol for women is 190 mg/dL. For 100 randomly selected women who immigrated in the past year, $\bar{x} = 178.2$ and $s = 45.3$.
(a) State suitable hypotheses.
(b) Test them at level 5%.
(c) Find the P-value of (b).
Solution.
(a) $H_0 : \mu = 190$, $H_1 : \mu \neq 190$.

(b) $t = \dfrac{178.2 - 190}{45.3/\sqrt{100}} = -2.60$. With $n = 100$ the t test applies without assuming normality, and $z_{0.025} = 1.96$ can replace the $t(99)$ quantile: the rejection region is $|t| \geq 1.96$. Since $|t| = 2.60 \geq 1.96$, H_0 is rejected at 5%.
(c) $P = P(|T| \geq 2.60) \approx 2 \times 0.0047 = 0.0094$ (normal table; software 0.0107): the mean differs significantly from the national mean.

6.2 Comparing means by paired comparison

Motivation. Applying both treatments to each unit (or matched pair) removes unit variation and reduces to a one-sample problem.

Comparing two means by observing pairs and analysing the within-pair differences is called **paired comparison**.

Illustration 6-1: two sleeping pills

Patients' health varies greatly, so each of 6 randomly chosen patients took both pills A and B once, in an order decided by a coin toss, and sleeping hours were recorded:

Patient	1	2	3	4	5	6
Pill A	4.8	4.0	5.8	4.9	5.3	7.4
Pill B	4.0	4.2	5.2	4.9	5.6	7.1

What can be inferred about $\mu_A - \mu_B$?

Illustration 6-2: wearable walking robot

To see whether a gait-assist robot increases exercise effect, 15 rehabilitation patients each walked 10 minutes with and without the robot; calories burned (kcal):

Patient	1	2	3	4	5	6	7	8
Without	23.9	23.9	20.0	19.2	22.2	11.8	21.7	27.7
With	25.2	28.2	24.9	20.2	19.7	16.9	22.6	20.8
Patient	9	10	11	12	13	14	15	
Without	15.6	22.5	23.2	19.3	15.2	10.0	11.2	
With	23.8	18.3	23.7	22.5	16.2	12.1	22.5	

The patients are **experimental units**, the pills or robot **treatments**, the change produced the **treatment effect**; pairing homogeneous units sharpens the comparison. With $(X_1, Y_1), \ldots, (X_n, Y_n)$, $\mu_1 = E(X_i)$, $\mu_2 = E(Y_i)$, form

$$D_i = X_i - Y_i, \qquad E(D_i) = \mu_1 - \mu_2, \qquad i = 1, \ldots, n,$$

and apply §6.1 to the sample $D_1, \ldots, D_n$ with mean $\bar{D}$ and standard deviation S_D.

Definition (Paired comparison of two means)

(a) A $100(1-\alpha)\%$ confidence interval for $\mu_1 - \mu_2$ is

$$\Big(\bar{d} - t_{\alpha/2}(n-1)\frac{s_D}{\sqrt{n}},\ \bar{d} + t_{\alpha/2}(n-1)\frac{s_D}{\sqrt{n}}\Big).$$

(b) For $H_0 : \mu_1 - \mu_2 = \delta_0$ the test statistic is $T = \dfrac{\bar{D} - \delta_0}{S_D/\sqrt{n}}$; with observed value t:

Alternative	P-value	Rejection region at level α
$H_1 : \mu_1 - \mu_2 > \delta_0$	$P(T \geq t)$	$t \geq t_\alpha(n-1)$
$H_1 : \mu_1 - \mu_2 < \delta_0$	$P(T \leq t)$	$t \leq -t_\alpha(n-1)$
$H_1 : \mu_1 - \mu_2 \neq \delta_0$	$P(\lvert T\rvert \geq \lvert t\rvert)$	$\lvert t\rvert \geq t_{\alpha/2}(n-1)$

These are exact for a normal random sample $D_1, \ldots, D_n$ (Theorem 4-10) and approximate for large n; units should be chosen at random and the treatment order randomized (**randomization**).

Example 6-4 (Sleeping pills: confidence interval)

Using Illustration 6-1, find a 95% confidence interval for $\mu_A - \mu_B$.
Solution. The differences A−B are 0.8, −0.2, 0.6, 0.0, −0.3, 0.3, so $\bar{d} = 0.20$ and $s_D = 0.443$. With $t_{0.025}(5) = 2.571$,

$$\bar{d} \pm t_{0.025}(5)\frac{s_D}{\sqrt{n}} = 0.20 \pm 2.571 \times \frac{0.443}{\sqrt{6}} = 0.20 \pm 0.46,$$

so the 95% confidence interval is $(-0.26,\ 0.66)$.

A normal quantile plot can check normality here, but more important is randomization in planning and observation.

Example 6-5 (Walking robot: test)

Using Illustration 6-2:
(a) test at level 1% whether wearing the robot increases calorie consumption;
(b) find the P-value of (a).
Solution.
(a) With μ_1, μ_2 the mean calories without and with the robot, $H_0 : \mu_1 = \mu_2$, $H_1 : \mu_1 < \mu_2$. The $n = 15$ differences (without − with) are

$$-1.3,\ -4.3,\ -4.9,\ -1.0,\ 2.5,\ -5.1,\ -0.9,\ 6.9,\ -8.2,\ 4.2,\ -0.5,\ -3.2,\ -1.0,\ -2.1,\ -11.3,$$

with $\bar{d} = -2.01$ and $s_D = 4.58$, so $t = \dfrac{-2.01 - 0}{4.58/\sqrt{15}} = -1.70$. Since $t_{0.01}(14) = 2.624$, the rejection region is $t \leq -2.624$; as $-1.70 > -2.624$, H_0 is not rejected at 1%.
(b) $P = P(T \leq -1.70)$, $T \sim t(14)$: the table gives $0.05 < P < 0.1$, and software $P = 0.056$. At the 1% level there is no clear evidence that the robot increases calorie consumption.

A normal quantile plot of these differences supports the normality assumption.

6.3 Comparing means with two independent samples

Motivation. With two separate groups, inference on $\mu_1 - \mu_2$ uses the difference of independent sample means.

Definition (Data structure for two-sample comparison)

(a) The observations in each group are a random sample from that group's population.
(b) Observations in different groups are independent.

Random samples from two populations satisfy this automatically; when one pool of units receives two treatments, the units must be split into the groups at random (randomization).

Label the populations $1, 2$ with means μ_1, μ_2 and standard deviations σ_1, σ_2, and let $X_{11}, \ldots, X_{1n_1}$ and $X_{21}, \ldots, X_{2n_2}$ be the samples, with

$$\bar{X}_1 = \frac{X_{11} + \cdots + X_{1n_1}}{n_1}, \quad S_1 = \sqrt{\frac{1}{n_1 - 1}\sum_{j=1}^{n_1}(X_{1j} - \bar{X}_1)^2},$$

and $\bar{X}_2, S_2$ likewise. We estimate $\mu_1 - \mu_2$ by $\widehat{\mu_1 - \mu_2} = \bar{X}_1 - \bar{X}_2$. By independence,

$$\mathrm{Var}(\bar{X}_1 - \bar{X}_2) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}, \qquad \text{so} \qquad \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}}$$

is the standardized difference, which is exactly $N(0,1)$ for normal populations (Theorem 4-2). With σ_1^2, σ_2^2 unknown we use the studentized difference

$$T = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{S_1^2/n_1 + S_2^2/n_2}}.$$

6.3.1 Large samples

When n_1, n_2 are both large, $S_1^2 \approx \sigma_1^2$ and $S_2^2 \approx \sigma_2^2$, so T is approximately $N(0,1)$ by the CLT.

Definition (Two-sample comparison of means (large n_1, n_2))

(a) A $100(1-\alpha)\%$ confidence interval for $\mu_1 - \mu_2$ is

$$(\bar{x}_1 - \bar{x}_2) \pm z_{\alpha/2}\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}.$$

(b) For $H_0 : \mu_1 - \mu_2 = \delta_0$ use $T = \dfrac{(\bar{X}_1 - \bar{X}_2) - \delta_0}{\sqrt{S_1^2/n_1 + S_2^2/n_2}}$, $T \dot{\sim} N(0,1)$; with observed value t:

Alternative	P-value	Rejection region
$H_1 : \mu_1 - \mu_2 > \delta_0$	$P(T \geq t)$	$t \geq z_\alpha$
$H_1 : \mu_1 - \mu_2 < \delta_0$	$P(T \leq t)$	$t \leq -z_\alpha$
$H_1 : \mu_1 - \mu_2 \neq \delta_0$	$P(\lvert T\rvert \geq \lvert t\rvert)$	$\lvert t\rvert \geq z_{\alpha/2}$

Example 6-6 (Tennis rackets and tennis elbow)

A new racket should reduce elbow stress. Of 220 players, 100 were randomly assigned the new racket and 120 the old; the elbow force (kgf) in a forehand stroke was:

	n_i	$\bar{x}_i$	s_i
New racket	100	25.2	5.1
Old racket	120	27.9	5.6

(a) Test at level 5% whether the mean forces differ.
(b) Find the P-value of (a).
(c) Find a 95% confidence interval for the difference of means.
Solution.
(a) $H_0 : \mu_1 = \mu_2$, $H_1 : \mu_1 \neq \mu_2$. The samples are large, and

$$t = \frac{(25.2 - 27.9) - 0}{\sqrt{5.1^2/100 + 5.6^2/120}} = -3.74.$$

The rejection region is $|t| \geq z_{0.025} = 1.96$; since $3.74 \geq 1.96$, H_0 is rejected at 5%.
(b) $P = P(|T| \geq 3.74)$ with $T \dot{\sim} N(0,1)$: $P < 0.0006$ from the table, and $P = 0.00018$ by software.
(c) $(25.2 - 27.9) \pm 1.96\sqrt{5.1^2/100 + 5.6^2/120} = -2.7 \pm 1.42$, i.e. $(-4.12, -1.28)$.

6.3.2 Normal populations: Welch's approximation

Even when n_1, n_2 are not large, T is approximately a t distribution with suitably chosen degrees of freedom. For normal populations with $n_1, n_2 \geq 5$,

$$T = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{S_1^2/n_1 + S_2^2/n_2}} \dot{\sim} t(df), \qquad df = \frac{\left(\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}\right)^2}{\frac{1}{n_1 - 1}\left(\frac{S_1^2}{n_1}\right)^2 + \frac{1}{n_2 - 1}\left(\frac{S_2^2}{n_2}\right)^2}.$$

Definition (Two-sample comparison of means (normal populations))

(a) A $100(1-\alpha)\%$ confidence interval for $\mu_1 - \mu_2$ is

$$(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2}(df)\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}},$$

with df computed from s_1^2, s_2^2 by the formula above.
(b) For $H_0 : \mu_1 - \mu_2 = \delta_0$ use $T = \dfrac{(\bar{X}_1 - \bar{X}_2) - \delta_0}{\sqrt{S_1^2/n_1 + S_2^2/n_2}} \dot{\sim} t(df)$; with observed value t:

Alternative	P-value	Rejection region
$H_1 : \mu_1 - \mu_2 > \delta_0$	$P(T \geq t)$	$t \geq t_\alpha(df)$
$H_1 : \mu_1 - \mu_2 < \delta_0$	$P(T \leq t)$	$t \leq -t_\alpha(df)$
$H_1 : \mu_1 - \mu_2 \neq \delta_0$	$P(\|T\| \geq \|t\|)$	$\|t\| \geq t_{\alpha/2}(df)$

Normality matters little for moderate n_1, n_2 unless Q–Q plots contradict it. For non-integer df use the nearest tabulated value, interpolate, or use software.

Example 6-7 (Grocery spending offline vs. online)

Is grocery spending lower offline than online? 18 and 20 random people bought two weeks' groceries for a family of four offline and online (thousand won):

Offline	169	206	257	294	252	283	240	207	230	183
	298	269	256	277	300	126	318	184		
Online	289	334	278	268	336	438	388	388	394	394
	425	386	356	342	305	365	355	312	209	458

(a) Test at level 5% whether offline spending is lower.
(b) Find the P-value of (a).
Solution.
(a) $H_0 : \mu_1 = \mu_2$, $H_1 : \mu_1 < \mu_2$ (1 = offline, 2 = online). Summary: $n_1 = 18$, $\bar{x}_1 = 241.6$, $s_1 = 52.82$; $n_2 = 20$, $\bar{x}_2 = 351.0$, $s_2 = 61.92$. The Q–Q plots of both groups are close to lines. Then

$$t = \frac{(241.6 - 351.0) - 0}{\sqrt{52.82^2/18 + 61.92^2/20}} = -5.875, \qquad df = \frac{\left(\frac{52.82^2}{18} + \frac{61.92^2}{20}\right)^2}{\frac{1}{17}\left(\frac{52.82^2}{18}\right)^2 + \frac{1}{19}\left(\frac{61.92^2}{20}\right)^2} = 35.91.$$

The rejection region is $t \le -t_{0.05}(35.91)$, where $t_{0.05}(35.91) \approx 1.620$ by interpolation in the table (1.688 by software). In either case $t = -5.875$ lies in the rejection region, so H_0 is rejected at 5%.
(b) $P = P(T \le -5.875)$, $T \sim t(35.91)$; the table gives $P < 0.0005$. There is clear evidence that offline grocery spending is lower.

6.3.3 Normal populations with a common variance

If $\sigma_1^2 = \sigma_2^2 = \sigma^2$, both S_1^2 and S_2^2 estimate σ^2, and combining them is more efficient. Weighting by degrees of freedom gives the **pooled sample variance**

$$S_p^2 = \frac{n_1 - 1}{(n_1 - 1) + (n_2 - 1)} S_1^2 + \frac{n_2 - 1}{(n_1 - 1) + (n_2 - 1)} S_2^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2},$$

with $n_1 + n_2 - 2$ degrees of freedom (Theorem 4-9).

	Population 1	Population 2	Pooled
Variance	σ^2	σ^2	
Sample variance	S_1^2	S_2^2	S_p^2
Degrees of freedom	$n_1 - 1$	$n_2 - 1$	$n_1 + n_2 - 2$

Under a common variance the standardized difference is $\frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sigma\sqrt{1/n_1 + 1/n_2}}$; replacing σ by S_p,

$$T = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t(n_1 + n_2 - 2)$$

for normal populations, by Theorem 4-11.

Definition (Two-sample comparison (normal populations, common variance))

(a) A $100(1-\alpha)\%$ confidence interval for $\mu_1 - \mu_2$ is

$$(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2}(n_1+n_2-2)\, s_p \sqrt{\frac{1}{n_1}+\frac{1}{n_2}}.$$

(b) For $H_0 : \mu_1 - \mu_2 = \delta_0$ use $T = \dfrac{(\bar{X}_1 - \bar{X}_2) - \delta_0}{S_p\sqrt{1/n_1 + 1/n_2}}$; with observed value t:

Alternative	P-value	Rejection region
$H_1 : \mu_1 - \mu_2 > \delta_0$	$P(T \geq t)$	$t \geq t_\alpha(n_1+n_2-2)$
$H_1 : \mu_1 - \mu_2 < \delta_0$	$P(T \leq t)$	$t \leq -t_\alpha(n_1+n_2-2)$
$H_1 : \mu_1 - \mu_2 \neq \delta_0$	$P(\lvert T\rvert \geq \lvert t\rvert)$	$\lvert t\rvert \geq t_{\alpha/2}(n_1+n_2-2)$

This is the **pooled two-sample t test**; the previous one, with adjusted degrees of freedom and no common-variance assumption, is the **unpooled two-sample t test**.

Pooled or unpooled?

The pooled test needs equal variances and degrades badly otherwise, especially for very different n_1, n_2; since equal variances are hard to verify, also examine the unpooled test.

Example 6-8 (Semaglutide and weight loss)

Twelve similar volunteers were split at random into two groups of 6: one received weekly semaglutide injections, the other a placebo, for 12 weeks with diet and exercise controlled. Weight loss (kg):

Semaglutide	5.12	4.87	5.64	5.01	5.49	5.25
Placebo	3.62	3.98	4.14	3.25	4.42	3.60

(a) Test at level 5% whether the mean weight losses differ.
(b) Find the P-value of (a).
(c) Find a 95% confidence interval for the difference of means.
Solution.
(a) $H_0 : \mu_1 = \mu_2$, $H_1 : \mu_1 \neq \mu_2$. Summary: $n_1 = n_2 = 6$, $\bar{x}_1 = 5.23$, $s_1 = 0.29$; $\bar{x}_2 = 3.84$, $s_2 = 0.42$. The Q–Q plots support normality and the standard deviations are similar, so we assume a common variance:

$$s_p^2 = \frac{(6-1)\times 0.29^2 + (6-1)\times 0.42^2}{6+6-2} = 0.13025, \qquad s_p = \sqrt{0.13025} = 0.361,$$

$$t = \frac{(5.23-3.84)-0}{0.361\sqrt{\frac{1}{6}+\frac{1}{6}}} = 6.67.$$

The rejection region is $|t| \geq t_{0.025}(10) = 2.228$; since $6.67 > 2.228$, H_0 is rejected at 5%.
(b) $P = P(|T| \geq 6.67)$, $T \sim t(10)$: $P < 0.001$. There is strong evidence that the mean weight losses differ.
(c) $(5.23-3.84) \pm 2.228 \times 0.361 \times \sqrt{\frac{1}{6}+\frac{1}{6}} = 1.39 \pm 0.46$, i.e. (0.93, 1.85).

The two-sample t tests and intervals are not very sensitive to the normality assumption and are approximately valid for large samples. What really matters is that the experiment was randomized.

6.4 Inference about population variances

Motivation. Variability itself matters (parts of the right mean size can still be defective); normal variances use chi-square and F.

6.4.1 One variance

For a random sample of size n from $N(\mu, \sigma^2)$ with μ, σ unknown, Theorem 4-8 gives

$$(n-1)\frac{S^2}{\sigma^2} \sim \chi^2(n-1).$$

With the quantiles $\chi^2_{1-\alpha/2}(n-1)$ and $\chi^2_{\alpha/2}(n-1)$,

$$P\Big(\chi^2_{1-\alpha/2}(n-1) \le (n-1)\frac{S^2}{\sigma^2} \le \chi^2_{\alpha/2}(n-1)\Big) = 1-\alpha,$$

and solving the inequalities for σ^2,

$$P\Big((n-1)\frac{S^2}{\chi^2_{\alpha/2}(n-1)} \le \sigma^2 \le (n-1)\frac{S^2}{\chi^2_{1-\alpha/2}(n-1)}\Big) = 1-\alpha.$$

Definition (Confidence interval for σ^2 (normal population))

A $100(1-\alpha)\%$ confidence interval for σ^2 is

$$\Big((n-1)\frac{s^2}{\chi^2_{\alpha/2}(n-1)},\ (n-1)\frac{s^2}{\chi^2_{1-\alpha/2}(n-1)}\Big).$$

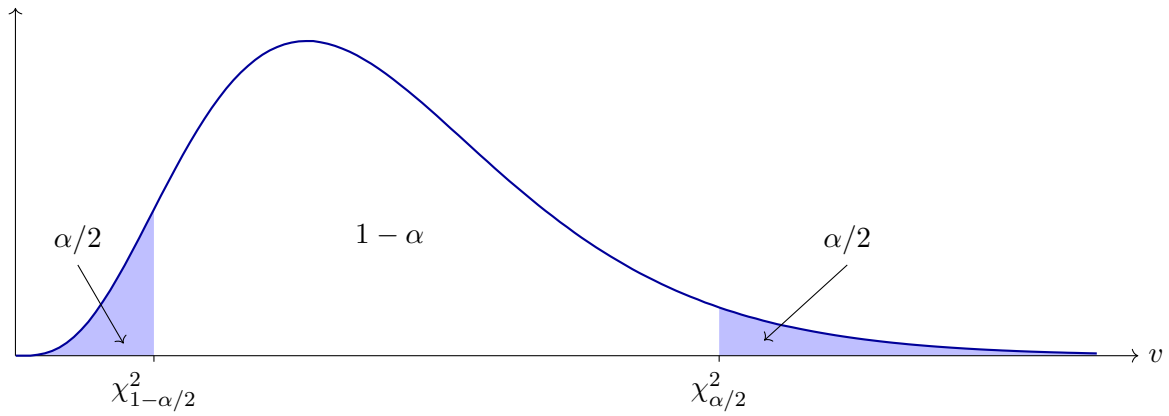


Figure 2: Quantiles of $\chi^2(9)$ used for a 90% interval for σ^2 ($\alpha = 0.1$): $\chi^2_{0.95}(9) = 3.33$, $\chi^2_{0.05}(9) = 16.92$.

The test of a variance based on the chi-square distribution is the **chi-square test**.

Definition (Chi-square test for σ^2 (normal population))

For $H_0 : \sigma^2 = \sigma_0^2$ use $\chi^2 = (n-1)\dfrac{S^2}{\sigma_0^2}$, which is $\chi^2(n-1)$ under H_0. With observed value χ_0^2:

Alternative	P-value	Rejection region at level α
$H_1:\sigma^2>\sigma_0^2$	$P(\chi^2\ge\chi_0^2)$	$\chi_0^2\ge\chi_\alpha^2(n-1)$
$H_1:\sigma^2<\sigma_0^2$	$P(\chi^2\le\chi_0^2)$	$\chi_0^2\le\chi_{1-\alpha}^2(n-1)$
$H_1:\sigma^2\ne\sigma_0^2$	$2P(\chi^2\ge\chi_0^2)$ or $2P(\chi^2\le\chi_0^2)$	$\chi_0^2\le\chi_{1-\alpha/2}^2(n-1)$ or $\chi_0^2\ge\chi_{\alpha/2}^2(n-1)$

For the two-sided alternative, compute both $2P(\chi^2\ge\chi_0^2)$ and $2P(\chi^2\le\chi_0^2)$ and use the one smaller than 1.

Example 6-9 (Coffee grinder)

A café uses 10 g of beans per Americano, but the grinder dose varies; a standard deviation above 1.5 g is considered abnormal. Ten doses (g):

11.15	9.62	9.83	10.85	12.66	9.51	11.23	9.38	9.11	8.56

Past records indicate a normal distribution.
(a) Test at level 5% whether the grinder is abnormal.
(b) Find the P-value of (a).
(c) Find a 90% confidence interval for the standard deviation of the dose.
Solution.
(a) $H_0:\sigma^2=1.5^2$, $H_1:\sigma^2>1.5^2$. Here $\bar{x}=10.19$, $s=1.24$, and

$$\chi_0^2=(n-1)\frac{s^2}{\sigma_0^2}=9\times\frac{1.24^2}{1.5^2}=6.15.$$

The rejection region is $\chi_0^2\ge\chi_{0.05}^2(9)=16.92$; since $6.15<16.92$, H_0 is not rejected at 5%.
(b) $P=P(\chi^2\ge6.15)$, $\chi^2\sim\chi^2(9)$; software gives $P=0.72$. There is no clear evidence of abnormal variation.
(c) With $\chi_{0.05}^2(9)=16.92$ and $\chi_{0.95}^2(9)=3.33$, the 90% interval for σ^2 is

$$\Big(9\times\frac{1.24^2}{16.92},\ 9\times\frac{1.24^2}{3.33}\Big)=(0.818,\ 4.156),$$

so the interval for σ is $(\sqrt{0.818},\sqrt{4.156})=(0.90,\ 2.04)$.

Sensitivity to normality

Unlike t methods, chi-square inference on a variance depends heavily on normality; check it first (the Q–Q plot for Example 6-9 is roughly linear).

6.4.2 Ratio of two variances

For independent random samples $X_{11},\dots,X_{1n_1}$ and $X_{21},\dots,X_{2n_2}$ from normal populations, with sample variances S_1^2,S_2^2, Theorem 4-13 gives

$$\frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2}\sim F(n_1-1,n_2-1).$$

Therefore

$$P\Big(F_{1-\alpha/2}(n_1-1,n_2-1)\le\frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2}\le F_{\alpha/2}(n_1-1,n_2-1)\Big)=1-\alpha,$$

and solving for σ_1^2/σ_2^2,

$$P\Big(\frac{S_1^2}{S_2^2}\cdot\frac{1}{F_{\alpha/2}(n_1-1,n_2-1)}\le\frac{\sigma_1^2}{\sigma_2^2}\le\frac{S_1^2}{S_2^2}\cdot\frac{1}{F_{1-\alpha/2}(n_1-1,n_2-1)}\Big)=1-\alpha.$$

By Theorem 4-12, $F_{1-\alpha/2}(n_1-1,n_2-1)=1/F_{\alpha/2}(n_2-1,n_1-1)$.

Definition (Confidence interval for σ_1^2/σ_2^2 (normal populations))

A $100(1-\alpha)\%$ confidence interval for σ_1^2/σ_2^2 is

$$\Big(\frac{s_1^2}{s_2^2}\times\frac{1}{F_{\alpha/2}(n_1-1,n_2-1)},\ \frac{s_1^2}{s_2^2}\times F_{\alpha/2}(n_2-1,n_1-1)\Big).$$

The test of σ_1^2/σ_2^2 based on the F distribution is the F **test**.

Definition (F test for the ratio of variances (normal populations))

For $H_0:\sigma_1^2=\sigma_2^2$ use $F=S_1^2/S_2^2$, which is $F(n_1-1,n_2-1)$ under H_0. With observed value f:

Alternative	P-value	Rejection region at level α
$H_1:\sigma_1^2>\sigma_2^2$	$P(F\ge f)$	$f\ge F_{\alpha}(n_1-1,n_2-1)$
$H_1:\sigma_1^2<\sigma_2^2$	$P(F\le f)$	$f\le 1/F_{\alpha}(n_2-1,n_1-1)$
$H_1:\sigma_1^2\ne\sigma_2^2$	$2P(F\ge f)$ or $2P(F\le f)$	$f\le 1/F_{\alpha/2}(n_2-1,n_1-1)$ or $f\ge F_{\alpha/2}(n_1-1,n_2-1)$

Example 6-10 (Polymer injection pressure)

Cracks in concrete are repaired by injecting a polymer. Injection pressures (10^6 psi) for two polymers:

Epoxy	1.75	2.12	2.05	1.97
MMA prepolymer	1.77	1.59	1.70	1.69

(a) Test at level 5% whether the variabilities of the injection pressures differ.
(b) Find a 95% confidence interval for the ratio of standard deviations.
Solution.
(a) $H_0:\sigma_1^2=\sigma_2^2$, $H_1:\sigma_1^2\ne\sigma_2^2$. Summary: $n_1=n_2=4$, $\bar{x}_1=1.97$, $s_1=0.160$; $\bar{x}_2=1.69$, $s_2=0.074$. The Q–Q plots do not contradict normality. The observed statistic is $f=0.160^2/0.074^2=4.675$. Since $F_{0.025}(3,3)=15.44$, the 5% rejection region is $f\le 1/15.44$ or $f\ge 15.44$. As $1/15.44<4.675<15.44$, H_0 is not rejected at 5%.
(b) The 95% interval for σ_1^2/σ_2^2 is

$$\Big(\frac{0.160^2}{0.074^2}\times\frac{1}{15.44},\ \frac{0.160^2}{0.074^2}\times 15.44\Big)=(0.301,\ 72.182),$$

so that for σ_1/σ_2 is $(\sqrt{0.301},\sqrt{72.182})=(0.549,\ 8.496)$.